

Inferential Statistics

Inferential statistics is a powerful extension of data analysis that allows us to move beyond describing observed data to making informed generalisations about a larger population. Rather than simply summarising a data set, it equips us with the tools to test assumptions, quantify uncertainty and evaluate claims using probability-based reasoning. By leveraging the logic of sampling and probability, inferential statistics helps data analysts determine whether patterns in data are meaningful or likely the result of random variation. This forms the backbone of rigorous, data-driven decision-making in research, business and scientific inquiry.

In this module, we will cover the following topics:

1. Central Limit Theorem
2. Hypothesis Testing – I
3. Hypothesis Testing – II

Central Limit Theorem

When working with real-world data, we often do not have access to the data of the entire population. Instead, we rely on samples to make estimates and decisions. However, samples are error-prone and may not reflect the population exactly.

We will discuss how sampling distributions help us study the variability across different samples. We will explore the shape, centre and spread of sampling distributions and learn how they impact the reliability of our estimates.

The central limit theorem (CLT) is a key principle that allows us to quantify the sampling error. It states that the sampling distribution of the sample mean tends to follow a normal distribution, even if the population itself is not normal, provided the sample size is large enough.

This result is powerful because it enables us to apply methods and inferences based on normal distribution in real-world scenarios where population distributions are unknown or non-normal.

Sample

In real-world data analysis, we often deal with a sample rather than the entire population. A sample is a subset of observations taken from the population and is used to estimate population characteristics. The central limit theorem builds on how these sample-based estimates behave when drawn repeatedly.

Population/Sample	Term	Notation	Formula
Population ($X_1, X_2, X_3, \dots, X_N$)	Population Size	N	Number of items/elements in the population
	Population Mean	μ	$\frac{\sum_{i=1}^N X_i}{N}$
	Population Variance	σ^2	$\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$
Sample ($X_1, X_2, X_3, \dots, X_n$) (Sample of Population)	Sample Size	n	Number of items/elements in the sample
	Sample Mean	$\bar{X}$	$\frac{\sum_{i=1}^n X_i}{n}$
	Sample Variance	s^2	$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$

Let's get acquainted with a few terms and phrases used in statistics and hypothesis testing:

- **Population size (N):**
The total number of elements or observations in the entire population
- **Population mean (μ):**
The average of all values in the population; represents the central tendency of the population
- **Population variance (σ^2):**
The average of the squared differences from the population mean; measures the overall spread or variability in the population data
- **Sample size (n):**
The number of elements or observations selected from the population to form a sample
- **Sample mean ($\bar{X}$):**
The average of the values in the sample; used as an estimate of the population mean
- **Sample variance (s^2):**
The average of the squared differences from the sample mean, divided by $(n - 1)$; used to measure variability in the sample and provides an unbiased estimate of population variance

These sample statistics form the foundation of the **sampling distribution**, which we will use to understand the behaviour of sample means, a crucial part of the central limit theorem.

Let's discuss an example of sampling.

To counter fake news, Facebook is planning to include a new feature in its timeline. Below each post, a fact-checking warning will be provided. This warning will notify users of a potentially false piece of news.

Before changing the timelines by including this feature for all Facebook users at once, Facebook first wants to evaluate how a few of its users will react to this new feature.

So, in a pilot launch, a small sample (~10,000 users) tries out the timeline with the new feature. Then, it asks the 10,000 users whether they prefer the new timeline (Feature B) or the old timeline (Feature A).

However, it is important to note that conclusions drawn from a sample do not always perfectly reflect the views of the entire population. Even though 50.5% of the sample preferred Feature B, this does not guarantee that the same proportion of the total user base will feel the same.

To understand how reliable this estimate is and how much it might vary from the true population preference, we need to study concepts such as sampling distributions, the central limit theorem and confidence intervals.

Let's say that one such survey shows that 50.5% of the people prefer Feature B to Feature A. Based on this, Facebook can say that more people prefer Feature B to Feature A; hence, B should replace A.

By conducting the exercise on a sample and not the entire population, it **saved time and money** and **avoided risks** that could arise if it rolled out an untested feature.

Sampling Distribution

Sampling distribution, specifically the sampling distribution of the sample means, is a probability distribution of the means calculated from all possible samples of a given size drawn from a population.

The mean of the sampling distribution is equal to the population mean (μ). This is because the average of all sample means converges to the true population mean.

Population/Sample	Term	Notation	Formula
Population ($X_1, X_2, X_3, \dots, X_N$)	Population Size	N	Number of items/elements in the population
	Population Mean	μ	$\frac{\sum_{i=1}^N X_i}{N}$
	Population Variance	σ^2	$\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$
Sample ($X_1, X_2, X_3, \dots, X_n$) (Sample of Population)	Sample Size	n	Number of items/elements in the sample
	Sample Mean	$\bar{X}$	$\frac{\sum_{i=1}^n X_i}{n}$
	Sample Variance	S^2	$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$
Sampling Distribution of the Sample Mean ($\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$) (k Sample Means)	Sampling Distribution's Size	No convention (We have used k, but that is not a norm)	
	Sampling Distribution's Mean (mean of sample means)	$\mu_{\bar{X}}$	$\mu_{\bar{X}} = \mu$
	Sampling Distribution's Standard Deviation	S.E. (Standard Error)	$S.E. = \sigma / \sqrt{n}$

Properties of Sampling Distribution of the Mean

- Sampling distribution's mean ($\mu_{\bar{x}}$) = Population mean (μ)
- Sampling distribution's standard deviation (Standard Error) = $\sigma / \sqrt{n}$
where σ is the population's standard deviation and n is the sample size
- For $n > 30$, the sampling distribution becomes a **normal distribution**

Estimating Mean Using Central Limit Theorem (CLT)

To estimate the population mean (μ) using a sample, we rely on the CLT, which allows us to treat the sampling distribution of the sample mean as approximately normal (for large enough sample sizes), with the following properties:

- Sampling distribution's mean:
 $\mu_{\bar{x}} = \mu$
- Sampling distribution's standard deviation (standard error):
 $\sigma/\sqrt{n}$ or approximately $S/\sqrt{n}$

Confidence Interval for the Population Mean

For a sample with:

- Sample size = n
- Sample mean = $\bar{X}$
- Sample standard deviation = S

The $y\%$ confidence interval for the population mean μ is given by:

$$\text{Confidence Interval} = (\bar{X} - Z^*S/\sqrt{n}, \bar{X} + Z^*S/\sqrt{n})$$

Where:

- Z^* is the Z-score corresponding to the $y\%$ confidence level.
- $S/\sqrt{n}$ is the standard error.
- The margin of error is $Z^*S/\sqrt{n}$.
- **Z-score** is the value corresponding to a specific **cumulative probability**.
- **Z^* (Z-star)** is the Z-score corresponding to a specific **confidence level**.

Hypothesis Testing – I

Hypothesis testing provides a formal, statistical framework to test assumptions about population parameters using sample data. Once you build a strong conceptual understanding of hypothesis testing, you will be able to apply it correctly in tools like Python, R or Excel.

Let's understand the basic difference between **inferential statistics** and **hypothesis testing**.

Inferential statistics is used to find a population parameter (mostly population mean) when you have no initial number to start with. So, you start with the sampling activity and find out the sample mean. Then, you estimate the population mean from the sample mean using the confidence interval.

Hypothesis testing is used to confirm your conclusion (or hypothesis) about the population parameter (which you know from exploratory data analysis (EDA) or your intuition). Through hypothesis testing, you can determine whether there is enough evidence to conclude if the hypothesis about the population parameter is true or not.

Hypothesis testing starts with the formulation of these two hypotheses:

- **Null hypothesis (H_0):**
This represents the current belief or the existing state of affairs — the status quo. We assume this to be true unless there is strong evidence against it. **The null hypothesis** is always represented by the following signs: $=$ OR $\leq$ OR $\geq$
- **Alternate Hypothesis (H_1):**
This represents a challenge to the status quo. It is what we want to test or prove using data. **The alternate hypothesis** is always represented by the following signs: $\neq$ OR $>$ OR $<$

Let's discuss the critical region and understand how the position of the critical region changes with the different types of null and alternate hypotheses.

The formulation of the null and alternate hypotheses determines the type of test and the position of the critical regions in the normal distribution.

You can tell the type of test and the position of the critical region based on the '**sign**' in the **alternate hypothesis**.

- $\neq$ in $H_1 \rightarrow$ Two-tailed test $\rightarrow$ Rejection region on **both sides** of distribution
- $<$ in $H_1 \rightarrow$ Lower-tailed test $\rightarrow$ Rejection region on **left side** of distribution
- $>$ in $H_1 \rightarrow$ Upper-tailed test $\rightarrow$ Rejection region on **right side** of distribution

The claim statement does not always directly represent the null hypothesis. To determine the null hypothesis, you need to examine whether the claim includes symbols like 'equal to', 'greater than or equal to', or 'less than or equal to'.

After formulating the hypothesis, you must follow these steps to **make a decision** using the **critical value method**:

1. Calculate the value of Z_c from the given value of α (significance level). Take it as 5% if not specified in the problem.
2. Calculate the critical values (UCV and LCV) from the value of Z_c .
3. Make the decision based on the value of the sample mean $\bar{x}$ with respect to the critical values (UCV and LCV).

Hypothesis Testing – II

The statistical methods in inferential statistics help you estimate a population mean using sample data, even without prior knowledge of the mean. However, in many real-world cases, you start with an assumption about the population and test whether the data supports or contradicts it. This is the purpose of hypothesis testing.

We have covered:

- The **p-value method** for hypothesis testing
- The **types of errors** in hypothesis testing

The **p-value method** is a widely used tool for statistical decision-making. You will learn how to interpret the p-value, relate it to significance levels and apply it in real-life scenarios.

We will also discuss **type-I** and **type-II errors**, which highlight the possible mistakes one can make based on data in hypothesis testing, and help evaluate the risks in statistical conclusions.

p- value

Let's get started with the p-value method of making a decision.

The higher the p-value, the higher the probability of failing to reject a null hypothesis. And the lower the p-value, the higher the probability of the null hypothesis being rejected.

After formulating the null and alternate hypotheses, we must follow these steps to **make a decision** using the **p-value method**:

1. Calculate the value of the z-score for the sample mean point on the distribution.
2. Calculate the p-value from the cumulative probability for the given z-score using the z-table.
3. Make a decision based on the p-value (multiply it by 2 for a two-tailed test) with respect to the given value of α (significance value).

To find the correct p-value from the z-score, find the **cumulative probability** first, by simply looking at the z-table, which will give you the area under the curve till that point.

A **p-value** is not the probability that the null hypothesis is true. Instead, it represents the probability of obtaining a result as extreme or more extreme than the observed one, assuming the null hypothesis is true. A high p-value indicates weak evidence against the null hypothesis but does not mean the null hypothesis is likely true.

When the **sample standard deviation (S)** is given and the population standard deviation (σ) is unknown, the **standard deviation of the sampling distribution (standard error)** is estimated using the following formula:

$$\text{Standard Error} = S / \sqrt{n}$$

This estimated standard error is then used in hypothesis testing and the critical value method.

While testing a hypothesis, you might make a wrong decision around accepting or rejecting the null hypothesis; such instances are referred to as 'errors'. Let's discuss the different types of errors in hypothesis testing.

Types of Errors

There are two types of errors that you might make in the hypothesis testing process:

1. Type-I error
2. Type-II error

	The null hypothesis is true	The null hypothesis is false
We decide to reject the null hypothesis	Type I error (rejecting a true null hypothesis) α	Correct decision
We fail to reject the null hypothesis	Correct decision	Type II error (failing to reject a false null hypothesis) β

A **type-I error**, represented by α , occurs when you reject a true null hypothesis.

A **type-II error**, represented by β , occurs when you fail to reject a false null hypothesis.

Let's take an example to understand the types of errors that can occur during hypothesis testing. Consider a medical test that diagnoses whether a person has COVID-19 or not.

Null Hypothesis (H_0):

The individual does not have COVID-19.

Alternate Hypothesis (H_1):

The individual has COVID-19.

Type-I Error (α)

A type I error occurs when the test result suggests that the person suffers from COVID-19, whereas the person does not have the infection. This is known as a false positive. It may lead to unnecessary isolation, anxiety or treatment.

Type-II Error (β)

A type-II two error occurs when the test result suggests that the person does not have COVID-19, whereas in reality, the person is infected. This is known as a false negative. It may result in the person going untreated and spreading the infection to others.

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